HARSHIT BHARDWAJ

Software Engineer (4+ YOE) | AI/GenAI Engineer

Python • FastAPI • MERN Stack • AWS • LLMs

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LinkedIn | GitHub

Technical Skills

Programming Languages: Python, NodeJs, ReactJs, JavaScript, TypeScript, SQL
AI / Generative AI: LLMs, Prompt Engineering, RAG, Agentic AI, LangChain, LangGraph, OpenAI, Claude, Embeddings, Vector Databases, Semantic Search, Tool Calling, Function Calling, Structured Output, Multi-Agent Systems, Document Processing, OCR
Backend Development: FastAPI, REST APIs, Microservices, Asynchronous Processing, Celery, WebSockets, Express.js
Databases & Storage: PostgreSQL, MongoDB, Redis, MySQL, DynamoDB, Elasticsearch
Cloud & DevOps: AWS (EC2, Lambda, API Gateway, S3, IAM, CloudWatch), GitHub Actions, PM2
Messaging & Distributed Systems: RabbitMQ, Kafka
Frontend: React.js, Next.js, Tailwind CSS, Material UI
Developer Tools: Git, GitHub

Experience

Software Engineer (AI Engineer) | Altre Digital Pvt. Ltd. *Aug 2024 – Present*

- **Architected** production-grade **Generative AI** applications using **Python**, **FastAPI**, **LangChain**, and enterprise LLMs to automate business workflows, reducing manual effort and improving operational efficiency.
- **Designed** an enterprise **multi-agent AI system** consisting of a **Supervisor Agent**, **8 specialized sub-agents**, and a **Humanized Answer Agent**, orchestrating complex workflows for office space planning, business location advisory, property intelligence, and enterprise automation.
- **Implemented** intelligent model routing across **GPT-4o**, **GPT-4.1**, **GPT-4.1 Mini**, **Claude Sonnet**, and **Claude Opus**, dynamically selecting the most suitable LLM based on reasoning complexity, latency, and task requirements.
- **Developed** AI agents with **tool calling**, custom REST API integrations, **Pydantic structured outputs**, prompt engineering, few-shot prompting, validation, retry mechanisms, and fallback strategies to deliver reliable enterprise AI workflows.
- **Built** an end-to-end **Retrieval-Augmented Generation (RAG)** platform for lease document intelligence using **OCRmyPDF**, semantic chunking, **text-embedding-3-small**, **Pinecone Vector Database**, semantic search, metadata-aware retrieval, and LLM-powered question answering over hundreds of scanned lease documents.
- **Implemented** scalable asynchronous AI document-processing pipelines using **FastAPI**, **Celery**, background workers, and **AWS S3 Presigned URLs**, automating document ingestion, OCR, embedding generation, vector indexing, and retrieval workflows.
- **Engineered** persistent conversational memory using **Redis** for chat history management and **WebSockets** for real-time streaming responses, enabling low-latency, context-aware AI interactions across enterprise applications.
- **Engineered** intelligent CAD/DXF parsing algorithms to automate workspace planning, desk allocation, occupancy analysis, meeting-room placement, and office layout optimization, reducing the overall layout design process from approximately **one week to nearly five minutes**.
- **Developed** an AI-powered Office Space Calculator by integrating multiple enterprise tools into a unified conversational interface, reducing manual operational effort by approximately **20%** through intelligent workflow automation.
- **Built** scalable backend microservices and reusable REST APIs supporting AI agents, workflow orchestration, authentication, document intelligence, and enterprise system integrations using modern microservice architecture.
- **Developed** interactive React.js(Typescript) applications using **React-Konva** for AI-assisted floor-plan visualization, real-time spatial editing, occupancy planning, and workspace optimization.
- **Led** end-to-end delivery of AI-enabled enterprise modules by collaborating with product managers and stakeholders, translating business requirements into scalable technical solutions, performing code reviews, and delivering production-ready releases.
- **Optimized** AI application performance through asynchronous processing, prompt optimization, semantic retrieval strategies, reusable architectures, and backend performance tuning, improving system responsiveness and developer productivity.

Software Engineer | Mediology Software *Jul 2022 – Aug 2024*

- **Designed** and developed scalable backend services and RESTful APIs using **Node.js**, **Express.js**, **Typescript**, and the **MERN** stack, supporting multiple enterprise applications with a focus on maintainability and performance.
- **Built** cloud-native backend solutions on **AWS** using **EC2**, **Lambda**, **API Gateway**, **DynamoDB**, **S3**, and **CloudWatch**, deploying scalable production applications with automated monitoring and logging.
- **Improved** API response latency from approximately **2 seconds** to **100–300 ms** through query optimization, efficient application logic, indexing strategies, and backend performance tuning.

- **Enhanced** application performance by approximately **30%** using **Redis** for distributed caching and **Elasticsearch** for high-performance full-text search and data retrieval.
- **Implemented** event-driven workflows using **Kafka** for asynchronous communication between distributed services, improving scalability and decoupling backend components.
- **Optimized** MongoDB performance through schema design, aggregation pipelines, indexing, and query optimization, improving application responsiveness under increasing workloads.
- **Managed** production deployments, server configuration, SSL certificate setup, monitoring, and application lifecycle management on Linux-based AWS environments, ensuring reliable and secure production releases.
- **Integrated** early Generative AI capabilities into backend services by experimenting with LLM APIs for intelligent workflow automation and internal productivity use cases.
- **Collaborated** with product managers, frontend engineers, QA teams, and DevOps engineers throughout the software development lifecycle, participating in technical design discussions, code reviews, production releases, and Agile delivery.

Education

B.Tech, Kurukshetra University 2019–2023
CGPA: 9.31

Projects

Enterprise Multi-Agent AI Platform *Python, FastAPI, LangChain, OpenAI, Claude, Pinecone*

- Built a production-grade multi-agent AI platform with a Supervisor Agent, 8 specialized AI agents, and Human Review workflow for enterprise automation.
- Developed end-to-end AI workflows using tool calling, Redis memory, WebSocket streaming, Pydantic structured outputs, and intelligent model routing across GPT-4o and Claude models.
- Integrated RAG capabilities using OCRmyPDF, semantic chunking, text-embedding-3-small, and Pinecone for intelligent document retrieval.

AI Workspace Planning & CAD Automation Platform

- Developed an AI-assisted workspace planning platform to automate DXF/CAD floor-plan analysis, desk allocation, occupancy planning, and office layout optimization.
- Reduced office layout planning time from approximately one week to nearly five minutes through intelligent automation and interactive visualization using React-Konva.

TheGameium – Gaming Tournament Platform thegameium.com

- Designed and developed a production-ready gaming tournament platform using the MERN stack with secure REST APIs, authentication, role-based access, and AWS deployment.
- Built scalable backend services supporting leaderboards, and scalable cloud infrastructure.

Publications

- **Getting Started with RabbitMQ in Node.js**
- **Adding TypeScript Support to an Existing React.js Project (Without Dropping JavaScript Support)**
- **Adding TypeScript Support to an Existing Node.js Project (Without Dropping JavaScript Support)**